

BROKILD · WINDOWS VST3 · DRUM KIT INSTRUMENT

BEETMACHINE

Eight slots, three drum synthesisers, one kit. No samples.

MANUAL · BUILD 260926.8

The image displays the BEETMACHINE VST3 interface, which consists of eight slots for drum synthesis and a detailed view of the Kickstart drum synthesiser.

Drum Slots:

- Slot 1:** KICKSTART, Studio Kick, C1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 2:** KICKSTART, Rock Kick, C#1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 3:** SNARE TACTICS, Studio Snare, D1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 4:** SNARE TACTICS, Rock Rim Shot, D#1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 5:** HATS OFF, Studio Hi-Hat, E1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 6:** HATS OFF, Loose Hat, F1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 7:** HATS OFF, Jazz Ride 20, F#1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.
- Slot 8:** HATS OFF, Crash 16, G1, LEVEL, PAN, CHOKED BY (1-8), MIX, OWN, BOTH.

Kickstart Drum Synthesiser:

- Header:** KICKSTART, BROKILD 260926.4, Studio Kick, SAVE, LOAD.
- Parameters:** PITCH 58.0 Hz, A#1 -8c, STARTS 84 Hz, 60 dB IN 380 ms, HIT, PITCH (Hz).
- Waveform:** A graph showing the pitch contour of the kick drum, with a peak at 58.0 Hz and a decay over time.
- Visuals:** A large circular speaker icon with the word "HIT" in the center.

WHAT IS IN HERE

03	ONE KIT, THREE DRUMS	What it is, and the first minute.
04	THE SLOT CARD	Everything on one card, top to bottom.
05	NOTES	C1, General MIDI, LEARN, layering.
06	CHOKES	One hit silencing another.
07	OUTPUTS & TIMING	MIX, OWN, BOTH, and why it lands on the beat.
08	THE DRUM PANELS	Each drum's own panel, inside.
09	THE HEADER	Kits, notes, master, BWFX, STOP, automation.
10	THE KITS	Twenty-four kits, every slot, and their racks.
12	THE BWFX RACK	FX and SPECTRA, order, macros, what a kit does to it.
13	SPECIFICATIONS	The numbers.

IN SHORT

Beetmachine is a drum kit. It has eight slots, and each one holds a real copy of one of three Brokild drum synthesisers — **Kickstart**, **Snare Tactics** or **Hats Off** — or nothing at all.

Click a slot and that drum's own full panel opens underneath. Nothing is sampled: every hit is synthesised as you play it.

THE FIRST TEN SECONDS

Open it and play C1 to G1 on a keyboard: that is the whole **STUDIO** kit. Then press > beside **KIT** and play them again.

ONE KIT, THREE DRUMS



The kit strip. The header, then eight slot cards. The STUDIO kit: two Kickstarts, two Snare Tactics, four Hats Off.

An occupied slot runs a **real copy** of that drum, built from the drum's own source — not a cut-down version of it. Whatever the drum can do on its own, it can do here.

An **empty slot costs nothing**: there is no drum in it to run. Eight empty slots measure 0.011 % of one core.

There is a convention, and it is not enforced: kicks in slots 1–2, snares in 3–4, hi-hats, ride, crash and splash in 5–8. Any slot can hold any drum.

1 • INSTALL

Copy **Beetmachine.vst3** into C:\Program Files\Common Files\VST3\ and rescan. The standalone needs no installing.

2 • PLAY

Slot 1 is on C1, slot 8 on G1. Or click the black **HIT** button on any card.

3 • EDIT

Click a card. That drum's own panel opens below the strip, with every control it has.

THE SLOT CARD



Slot 5, selected. The amber rim marks the card whose panel is open below.

NUMBER & TYPE

The slot's number, and what it holds: **EMPTY**, **KICKSTART**, **SNARE TACTICS** or **HATS OFF**.

PRESET < >

The drum's own factory presets. The arrows step through them; the name opens the list.

HIT & LAMP

The black palm button plays the slot. The amber jewel lamp beside it lights on every hit, from MIDI too. Then the slot's note.

- + LEARN

Note down or up a semitone. **LEARN**: press it, then play a key — that key now triggers this slot.

LEVEL & PAN

The amber mark on each knob's scale shows its setting. Double-click: **0 dB**, and **centre**.

M & S

Mute and solo. While any slot is soloed, only soloed slots sound.

CHOKED BY 1-8

Light another slot's number and a hit there silences this one. The card's own number is dimmed: a slot cannot choke itself. See page 6.

OUT

MIX, **OWN** or **BOTH**: where the slot's sound goes. See page 7.

NOTES

One note per slot. Two rows are built in; any slot can be moved.

SLOT	CONVENTION	C1 ROW	GM ROW
1	kick	C1 · 36	C1 · 36
2	kick	C#1 · 37	B0 · 35
3	snare	D1 · 38	D1 · 38
4	snare	D#1 · 39	E1 · 40
5	closed hat	E1 · 40	F#1 · 42
6	open hat	F1 · 41	A#1 · 46
7	ride	F#1 · 42	D#2 · 51
8	crash	G1 · 43	C#2 · 49

Note names follow Ableton Live, where C1 is MIDI note 36 and C3 is 60.

C1 & GM

C1 puts the kit on eight neighbouring keys from C1 up, where drum MIDI usually sits. **GM** sets the General MIDI drum notes, so a MIDI file or groove written for GM plays the right drums. Once any note has been moved, the header reads **CUSTOM**. A project saved on the old C3 row keeps its notes.

– + LEARN

Nudge a slot a semitone at a time, or press **LEARN** and play the key you want.

LAYERING

Two slots on the same note both play. Put a Kickstart and a Snare Tactics on one key and it is one hit.

KEYS STILL WORKS

Each drum’s own **KEYS** switch still works in its panel. Whatever it is set to, Beetmachine plays the sound dialled on that drum.

CHOKES



Slot 6, the open hat. The lit **5** in CHOKED BY: a hit on the closed hat silences it.

A real open hi-hat stops ringing the moment the pedal closes it. A kit without that sounds wrong straight away, and this is how Beetmachine does it.

CHOKED BY

On each card, light any other slot's number. A hit on that slot now silences this one, with a **3 ms** fade so it does not click. Light several numbers and any of them will do it.

AFTERWARDS

A choked slot is not switched off. It plays normally on its next hit.

IN THE KITS

In every kit the open hat in slot 6 is choked by the closed hat in slot 5. **LO-FI** also chokes its splash, slot 8, from the ride in slot 7.

TRY THIS

Light **2** on the crash in slot 8. Now a hit on the second kick grabs the crash — a stop hit, the way a drummer would catch the cymbal.

OUTPUTS & TIMING

Stereo is the default. Eight more stereo outputs, one per slot, are there when you want them.

OUT WHERE THE SLOT GOES

MIX The main stereo mix. Every slot starts here.

OWN Only the slot's own stereo output pair. It is taken out of the mix, so it is not heard twice.

BOTH The main mix and its own pair.

ENABLING AN OUTPUT

The eight extra pairs are switched off until the host enables one. Hosts list them as **Mix** and **Slot 1** to **Slot 8**; route one to a track in your DAW and it comes up.

OFF IN HOST

A slot set to OWN whose pair is off in the host stays in the mix rather than going silent, and its card says **OFF IN HOST**.

EVERY SLOT ON THE SAME SAMPLE

The three drums have different built-in latencies: at 48 kHz, Kickstart **95** samples, Snare Tactics **95**, Hats Off **103**.

Beetmachine delays the quicker ones to match, so drums played together land on the same sample, and it tells the host its latency so the host can compensate.

WHAT A KIT DOES NOT TOUCH

Loading a kit changes the sounds, levels, pans and chokes. It never changes your notes or your output routing.

THE DRUM PANELS



Slot 1 selected. Kickstart's own panel opens under the strip — the display, the HIT pad, every section.

CLICK A CARD

That drum's own panel opens under the strip — unchanged: every control, its preset browser, its SAVE and LOAD.

SAVED WITH THE PROJECT

Everything set in a drum's panel is kept with the project. It is not automated control by control.

ONE FAMILY

KICKSTART a red-lead steam pile-driver,
SNARE TACTICS olive-drab field gear,
HATS OFF a lathe in a cymbal foundry.



Slot 5 selected. Hats Off, with its cymbal pad and its CLOSED, PEDAL and OPEN buttons.

THE HEADER



The header. Nameplate, KIT, NOTES, MASTER, the build number and the red STOP.

KIT < >

Twenty-four themed kits. The arrows step through them; the name opens the list. The kits are on pages 10 and 11.

NOTES

C1 or **GM**, and **CUSTOM** once a note has been moved. See page 5.

MASTER

The red knob: the level of the whole kit.

BWFX

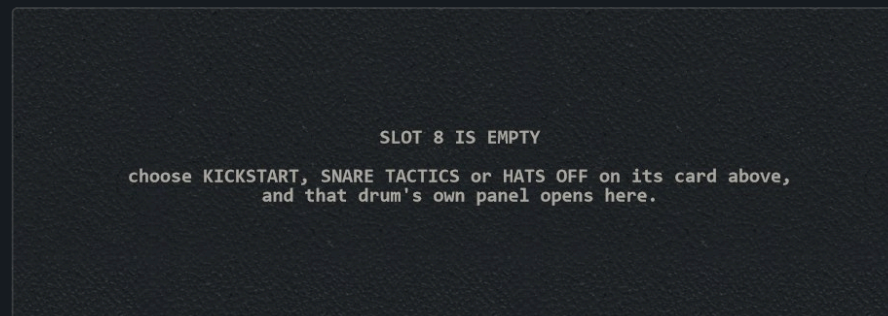
The Brokild World FX rack on the main mix: the same FX and SPECTRA racks every Brokild synth has. A kit sets it. A slot sent to its OWN output goes round it.

STOP

The red emergency stop. Every slot fades out at once.

AUTOMATION

The host sees each slot's **level**, **pan** and **mute**, **Master** and **BWFX MACRO 1–5**. The drums' own controls are saved with the project but not automated one by one.



An empty slot selected. No drum, no panel, no CPU. Choose a drum on its card and its panel opens here.

THE KITS

KIT	1 KICK	2 KICK	3 SNARE	4 SNARE	5 HAT	6 OPEN HAT	7	8
STUDIO	Studio Kick	Rock Kick	Studio Snare	Rock Rim Shot	Studio Hi-Hat 14	Loose Hat	Jazz Ride 20	Crash 16
808	808 Boom	808 Short	808 Snare	808 Clap	808 Closed Hat	808 Open Hat	808 Cymbal	606 Hats
909	909 Classic	909 Hard	909 Snare	909 Tight	909 Closed Hat	909 Open Hat	909 Ride	909 Crash
EIGHTIES	LinnDrum	DMX	Big Eighties	Simmons SDS-V	LinnDrum Hat	707 / 727 Hat	Ride Bell	Crash 18
BOOM BAP	SP-12 Boom Bap	MPC60 Thump	SP-1200 Crack	MPC60 Break	SP-1200 Hat	Loose Hat	Sizzle Ride	Crash 18
TECHNO	Techno Rumble	Hard Techno	Techno Crush	Warehouse Clap	Techno Closed	Rolling Open Hat	Detroit Ride	Minimal Tick
DUB	Clean Sub	Felt Beater	Steppers	Tubby Throw	Roots Hi-Hat	Dub Hat Echo	Skank Hat	Space Cymbal
INDUSTRIAL	Industrial Crush	Gabber	Metal Shop	EBM Snap	Hard Techno Hat	Kraftwerk Metal	Metal Plate	China 18
TRAP	Trap 808	Dubstep Punch	Trap Snare Clap	DnB Snare	Trap Hat	UKG Shuffle Hat	Grime Hat	Halftime Crash
LO-FI	Lo-Fi Dust	Towel In The Drum	Lo-Fi Ghost	Seventies Dead	Lo-Fi Hat	Hats In A Room	Dark Ride 22	Splash 10
DUB SPACE SPACE SPRING	Clean Sub	Felt Beater	Roots Rim Shot	Steppers	Roots Hi-Hat	Dub Hat Echo	Skank Hat	Space Cymbal
DNB ROLLER STRIP	909 Classic	Clean Sub	DnB Snare	Amen Chop	Studio Hi-Hat 14	Loose Hat	Sizzle Ride	Crash 18

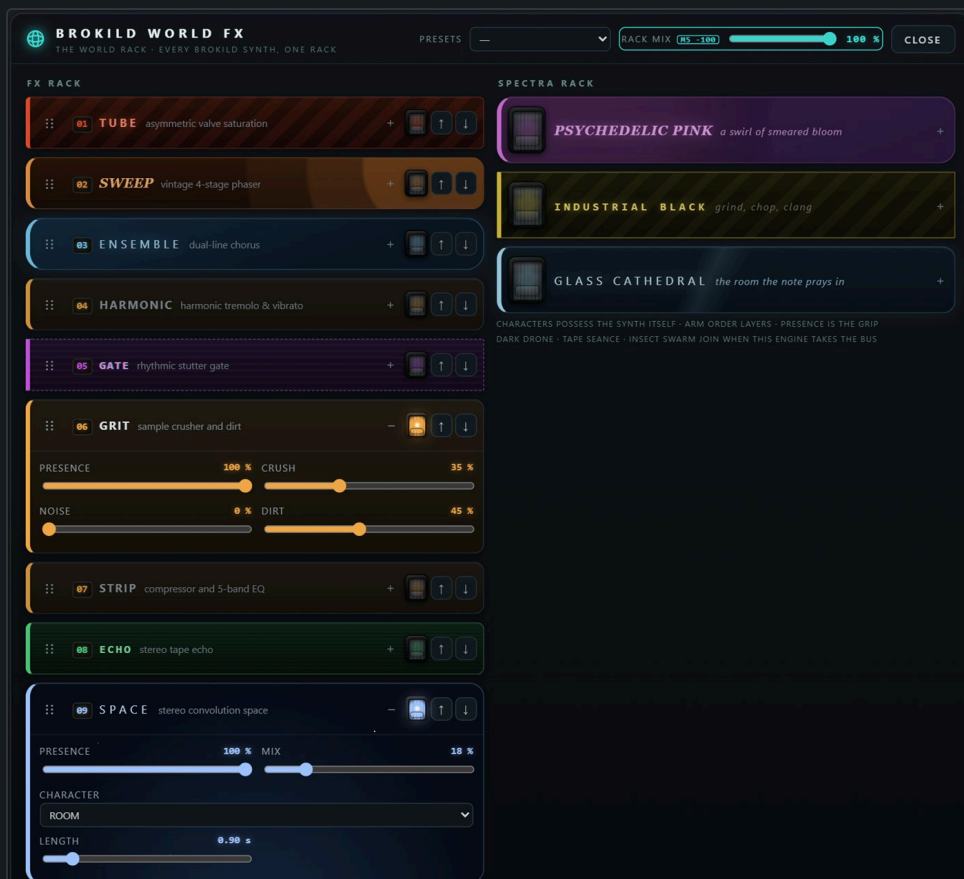
Slots 1–2 are Kickstart, 3–4 Snare Tactics and 5–8 Hats Off in every kit, and every name is one of that drum’s own presets. The open hat in slot 6 is choked by the closed hat in slot 5; in LO-FI the splash is also choked by the ride.

THE KITS, CONTINUED

KIT	1 KICK	2 KICK	3 SNARE	4 SNARE	5 HAT	6 OPEN HAT	7	8
DNB NEURO GRIT + STRIP	Dubstep Punch	Hard Techno	DnB Snare	Metal Shop	Hard Techno Hat	Rolling Open Hat	Metal Plate	China 18
AMEN BREAKS GRIT + STRIP	MPC60 Thump	Kit In A Room	Amen Chop	MPC60 Break	SP-1200 Hat	Loose Hat	Jazz Ride 20	Crash 16
BIG BEAT STRIP + SPACE ROOM	Rock Kick	Studio Kick	Piccolo Funk	Loose Garage	Studio Hi-Hat 14	Loose Hat	Sizzle Ride	Crash 18
DOOM SPACE HALL	Rock Kick	Kit In A Room	Big Eighties	Rock Rim Shot	Loose Hat	Hats In A Room	Dark Ride 22	China 18
BLACK METAL	Studio Kick	Studio Kick	Piccolo Funk	Rock Rim Shot	Studio Hi-Hat 14	Loose Hat	China 18	Crash 18
FACTORY GRIT + SPACE ROOM	Industrial Crush	Techno Rumble	Factory Floor	Power Electronics	Kraftwerk Metal	Metal Plate	Warehouse Ride	China 18
EBM STRIP + SPACE PLATE	Drum Computer	Hard Techno	EBM Snap	Gated Machine	Techno Closed	Rolling Open Hat	Kraftwerk Metal	Crash 16
JAZZ CLUB SPACE ROOM	Jazz Kick	Felt Beater	Jazz Snare	Seventies Dead	Studio Hi-Hat 14	Loose Hat	Jazz Ride 20	Sizzle Ride
IDM GRIT + ECHO	808 Short	KR-55	Simmons SDS-V	Minimal Rim	Minimal Tick	606 Hats	CR-78 Hat	Splash 10
DOWNTEMPO GRIT + SPACE PLATE	CR-78	Felt Beater	CR-78	Space Echo Clap	CR-78 Hat	Hats In A Room	606 Hats	Space Cymbal
DUB TECHNO SPACE HALL	House Thump	Techno Rumble	Dub Techno Space	Minimal Rim	Dub Techno Hat	Rolling Open Hat	Detroit Ride	Space Cymbal
HARD TECHNO TUBE + STRIP	Hard Techno	Techno Rumble	Techno Crush	Warehouse Clap	Hard Techno Hat	Rolling Open Hat	Warehouse Ride	Crash 18

A kit sets sounds, levels, pans, chokes and the **BWFX rack** — never notes or routing. The thirteen kits with an effect named under them bring their own rack, and every other kit loads with an empty one, the way a patch sets the rack on every Brokild synth. The later kits also turn a few of each drum’s own controls after its preset loads: BLACK METAL’s two kicks are one drum, short and clicky, tuned a hair apart — the two feet of a double pedal.

THE BWFX RACK



The **FACTORY kit's rack**. GRIT and a SPACE room armed; every other pedal waits, folded.

ONE RACK, EVERY SYNTH

The **BWFX** button opens the same Brokild World FX rack every Brokild synth has: FX pedals on the left, SPECTRA characters on the right, the five macros underneath.

ORDER

Drag a pedal by its grip, or use its ↑ ↓: the chain runs top to bottom, so a crusher before a room is a different sound from a room before a crusher.

A KIT SETS IT

Twenty-four kits, and thirteen bring a rack of their own; the rest load with an empty one. Your project remembers the rack as you left it.

ON THE MIX ONLY

The rack sits on the main stereo mix. A slot sent to its OWN output goes round it; BOTH sends the rack its copy and the output a dry one.

MACROS

BWFX MACRO 1–5 are host parameters, so a DAW can automate the rack. Macro 5 is the rack's dry/wet.

SPECIFICATIONS

Format	Windows VST3, 64-bit, and a standalone application
Type	Instrument / Drum. MIDI in; a stereo mix plus eight stereo slot outputs
Sound	Fully synthesised. No samples
Slots	Eight. Each empty, Kickstart, Snare Tactics or Hats Off
Kits	24 themed kits, 13 with their own BWFX rack
Notes	C1 row, General MIDI row, or any note per slot
Host parameters	Level, pan and mute per slot, Master, BWFX macros 1–5
Licence	AGPLv3. Source at github.com/peterboggild/BrokildApps
Price	Free
Build	260926.8

MEASURED

Eight empty slots	0.011 % of one core
Full kit, all eight hit on every 16th	40–46 % of one core
Choke fade	3 ms
Latency at 48 kHz	103 samples
Kickstart · Snare Tactics · Hats Off	95 · 95 · 103

The CPU of a full kit is the three drums' own cost, not Beetmachine's. Every slot is aligned to the slowest drum, and the latency is reported to the host.

CREDITS

NOW GO AND HIT SOMETHING

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Cheers, Peter Bøggild, September 2026.

Drum machine names are trademarks of their owners, used only to describe the sound a preset is in the spirit of. Beetmachine contains no samples of them.

BROKILD

Free, and free to pass on.
peterboggild.github.io/BrokildApps

BUILD 260926.8 · WINDOWS VST3

